

Vedant Kulkarni

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EDUCATION

- Indian Institute of Information Technology, Lucknow** India
Bachelor of Technology — Computer Science and Business; CGPA: 8.62 Aug-2024 – May-2028

SKILLS

- Programming Languages:** Go, C++, JavaScript, TypeScript, Python
- Web Development:** Node.js, Express.js, React.js, Next.js, Tailwind CSS, REST APIs, WebSockets, JWT, Redis
- Databases:** PostgreSQL, MongoDB
- Cloud& DevOps:** Git, Docker, Vercel, Render
- CS Fundamentals:** Data Structures and Algorithms, Database Management Systems, Object Oriented Programming, Operating System

TECHNICAL PROJECTS

- Sentinel — Distributed Rate Limiting System (Go) — [GitHub] [Live]: Tech Stack:** Go, Redis, REST APIs, Goroutines, Mutex/Atomic Operations
 - Architected a distributed rate-limiting system in Go (Token Bucket and Sliding Window algorithms), capable of handling 50,000+ RPS per node with a 2ms p99 latency overhead by utilizing a hybrid local burst and global Redis architecture.
 - Optimized high-concurrency request processing leveraging goroutines and atomic operations; implemented a thread-safe cleanup sweeper for expired client keys and a Redis shared connection pool to guarantee 100% fail-open resiliency.
 - Engineered HTTP middleware featuring longest-prefix route matching to secure nested subpaths, and integrated a path-labeled Prometheus exporter to track allowed and blocked requests globally with 1ms precision.
- Opensource Compass — AI-Powered Open Source Discovery Platform — [GitHub] [Live]: Tech Stack:** Next.js 15, TypeScript, Go, Supabase (PostgreSQL), WebSockets, Docker Compose, Groq (Mixtral), Google Gemini, GitHub OAuth, JWT
 - Architected a microservices-based platform with 5 independent Go services (Auth, GitHub Proxy, AI, Core/Orchestration, Notifications) orchestrated via Docker Compose, implementing a multi-signal scoring engine to rank GitHub repositories (0–100) based on issue accessibility, maintainer responsiveness, documentation quality, and tech stack compatibility.
 - Designed an AI-powered enrichment pipeline using provider-agnostic LLM abstractions with Groq (Mixtral) and Gemini fallback to generate contribution guides and estimate issue difficulty for first-time contributors.
 - Built a concurrent real-time notification infrastructure using Go WebSockets, per-user connection management, and GitHub OAuth authentication, enabling personalized repository watchlists and activity tracking.
 - Led development during a college-wide FOSS event, onboarding 10+ contributors and maintaining contribution workflows through issue triaging, code reviews, and repository documentation.
- BawarchiAI — AI-Driven Food Waste Reduction Platform — [GitHub] [Live]: Tech Stack:** Next.js, Go (Gin), PostgreSQL (Supabase), Python, FastAPI, XGBoost, REST APIs, Docker
 - Architected a scalable microservices backend in Go (Gin) and a containerized FastAPI ML service, integrating with Supabase PostgreSQL via PgBouncer transaction poolers for optimized serverless database routing.
 - Engineered a hybrid forecasting algorithm blending a 5-Fold Cross-Validation XGBoost regressor—achieving a 97.9% out-of-sample generalization (R^2)—with historical baselines to deliver real-time recommendations with 350ms inference latency.
 - Developed responsive Next.js dashboards featuring real-time canteen telemetry and feature importance visualizations, enabling seamless surplus acceptance and NGO redistribution logs.

ACHIEVEMENTS

- Winner - DevMatrix** (Open-Domain Hackathon) for building CampusEx, a gamified real-time social stock market featuring peer trading and an AI-driven news engine for dynamic price volatility.
- Winner - Reckon 7.0** (GenAI Hackathon) for building an AI-powered investment intelligence platform
- Achieved a peak rating of **1336 (Pupil)** at Codeforces Round 1107 (Div. 3) — **Global Rank 2001** Ranklist Link
- Global Rank 3530** at Codeforces Pinely Round 5 (Div. 1 + Div. 2) [Ranklist] — [Codeforces]
- Global Rank 471** at CodeChef Starters 209 [Contest] — [CodeChef]

POSITION OF RESPONSIBILITY

- Member, Web Wing — Axios (Technical Club), IIIT Lucknow Sept-2025 – Present
- Member, Seraphim (Social Awareness Club), IIIT Lucknow Sept-2024 – Present
- Member, Zephyr (Dance Society), IIIT Lucknow Sept-2024 – Present